5.

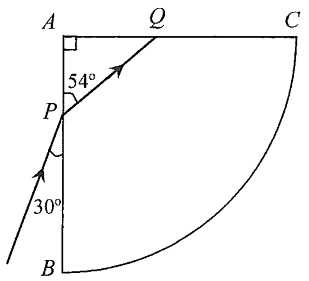

Figure 5.1

Figure 5.1 shows the cross-section of a glass block *ABC*. *ABC* is a quarter circle with its centre at *A*. A ray of red light is incident at *P* on face *AB* and the refracted ray strikes the face *AC* at *Q* as shown.

(a) Calculate the refractive index of the glass for red light. (2 marks)

(b) Explain why the ray is totally reflected when it strikes the face *AC* at *Q*. (2 marks)

(c) In Figure 5.1, sketch the subsequent path of the ray until it finally emerges from the block to the air. (2 marks)

(d) If the incident ray is a ray of white light, what can be observed when it finally emerges from the block ? (1 mark)

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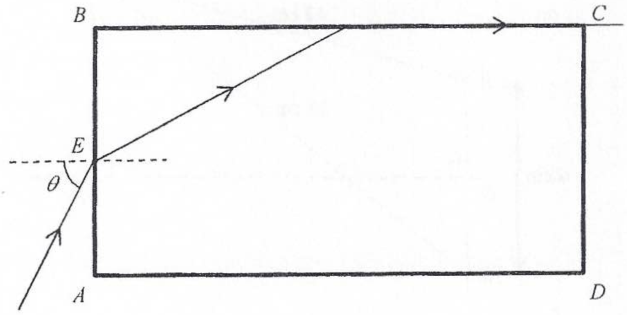
7. (a) A light ray enters a rectangular plastic block $ABCD$ from air at point E, and the angle of incidence is θ. The light ray emerges along face BC as shown in Figure 7.1. The refractive index of the plastic is 1.36.

Figure 7.1



(i) Find the critical angle of the plastic. (2 marks)

(ii) Find the value of θ. (3 marks)

(iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ, sketch the path of the light ray in Figure 7.1. (2 marks)

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Answers written in the margins will not be marked.

(b) A student designs a periscope using two plastic prisms, the refractive index of the plastic is 1.36. As shown in Figure 7.2, an object is placed in front of the periscope.

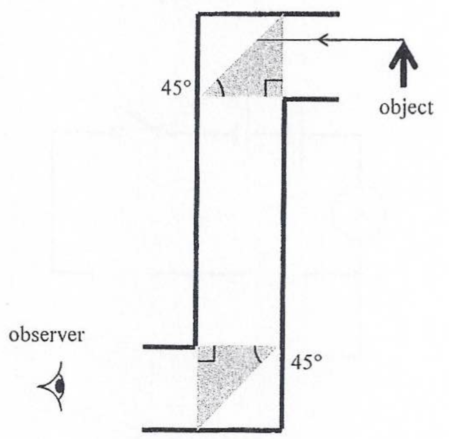

Figure 7.2

(i) Complete the path of the light ray from the object in Figure 7.2, and explain why the periscope fails to work. (3 marks)

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(ii) What can be used to replace the two plastic prisms so that the periscope can work properly ? (1 mark)

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7. Figure 7.1 shows an optical fibre which consists of a cylindrical glass core of refractive index n_g enclosed by a transparent cladding of refractive index n_c.

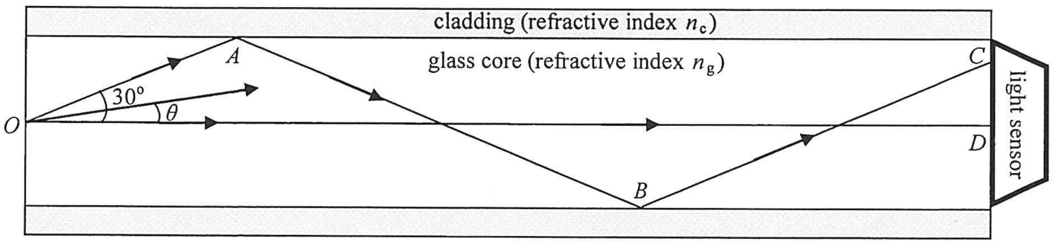

Figure 7.1

As shown in Figure 7.1, a point light source at O emits monochromatic light in all directions. Inside the fibre, light can reach the right end of the fibre through many different paths making angles θ with the axis OD. Two of these paths, OD and $OABC$, have been drawn for reference. Light ray OA makes an angle of 30° with the axis OD and is incident at the core-cladding boundary at A with an angle of incidence i_A.

(a) (i) Find i_A. (1 mark)

(ii) If i_A is just greater than the critical angle of that boundary, estimate $\dfrac{n_g}{n_c}$. (2 marks)

(iii) What phenomenon occurs at point A ? State the condition needs to be satisfied by θ such that this phenomenon **fails to occur**. (2 marks)

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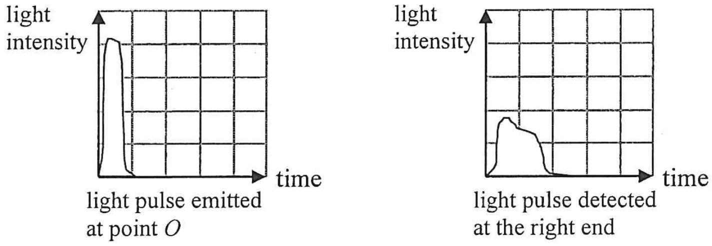
(b) A narrow monochromatic light pulse (i.e. of a short duration) emitted at point O propagates with its energy within $\theta = \pm 30°$ towards a light sensor located at the right end of the optical fibre. The respective emitted and detected light pulses are represented below using the same scales.

Figure 7.2



(i) Explain why the light pulse detected is **broader** (i.e. of a longer duration) and with **lower intensity**. Assume that the loss of energy of the light pulse due to absorption by glass is negligible. (2 marks)

(ii) An engineer suggests changing the refractive index n_c of the cladding in order to reduce the width of the light pulse received. Should n_c be increased or decreased ? Or, will a change in n_c have no effect on the pulse width ? Explain your choice. (2 marks)

6. Read the following passage about **rainbows** and answer the questions that follow.

Rainbows are sometimes seen after a rainfall (Figure 6.1(a)). Tiny water droplets in the air each act like a prism to disperse sunlight into a spectrum of colours. The most common rainbow that we see is called a primary rainbow, in which the white light from the sun undergoes reflection once within each water droplet.

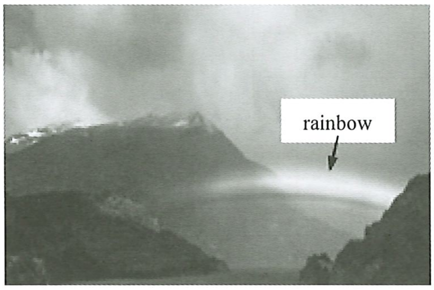


Figure 6.1(a)

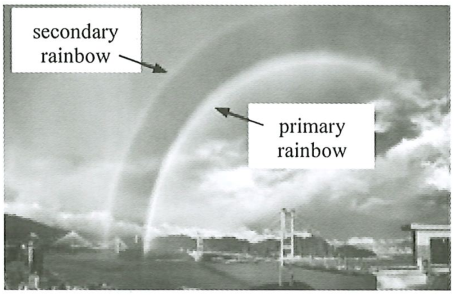


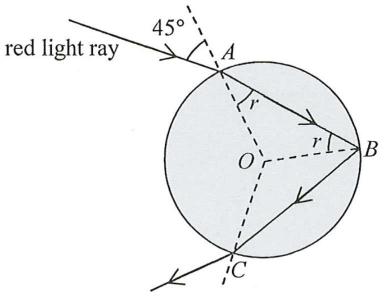
Figure 6.1(b)

Under suitable conditions, a double rainbow with an additional secondary rainbow formed above the primary rainbow can be seen (Figure 6.1(b)).

Assume that water droplets are spherical. Given: refractive index for red light in water = 1.325

(a) Figure 6.2 shows a red light ray entering a water droplet suspended in air.

Figure 6.2



(i) When the angle of incidence is 45°, find the angle of refraction r at point A. (2 marks)

(ii) Calculate the critical angle c for red light at the water-air interface and justify whether the reflection at point B is a total internal reflection or not. (3 marks)

Answers written in the margins will not be marked.

(b) (i) Figure 6.3 illustrates how an observer sees a rainbow. O_1 and O_2 are the respective centres of the upper and lower water droplets suspended in air. The two paths of red light, R_1 and R_2, in the white light from the Sun are drawn for your reference, with R_1 being seen by the stationary observer.

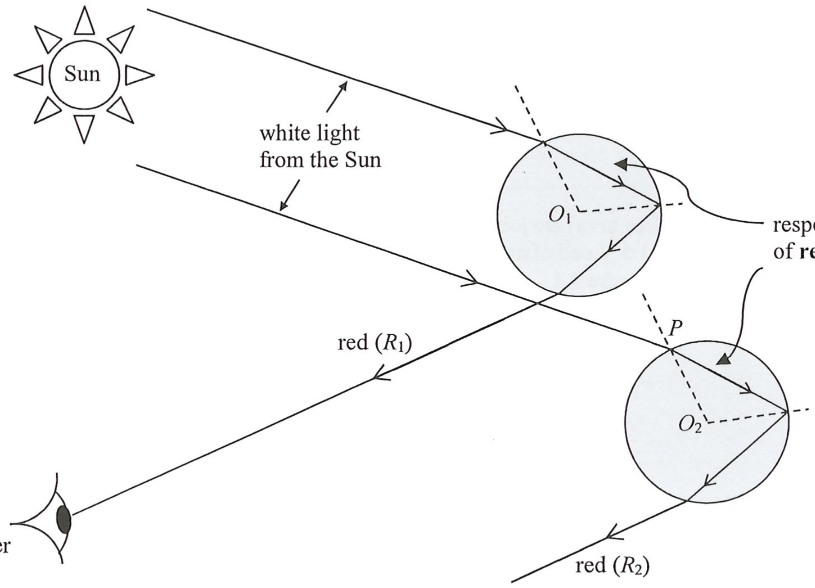


Figure 6.3

The lower water droplet enables **a violet light ray** from sunlight to be seen by the observer. Sketch the path of this violet light ray starting from point P on the lower water droplet. Given that the refractive index for violet light in water is slightly greater than that for red light. (2 marks)

(ii) For a secondary rainbow to be formed, a light ray has to undergo ***reflection twice within each of those water droplets*** situated higher above as shown in Figure 6.4.

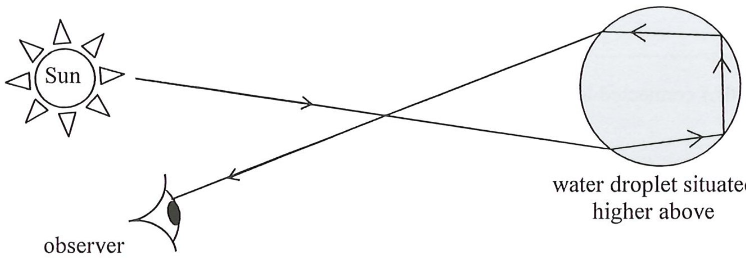


Figure 6.4

Explain why a secondary rainbow is dimmer than the primary rainbow. (2 marks)

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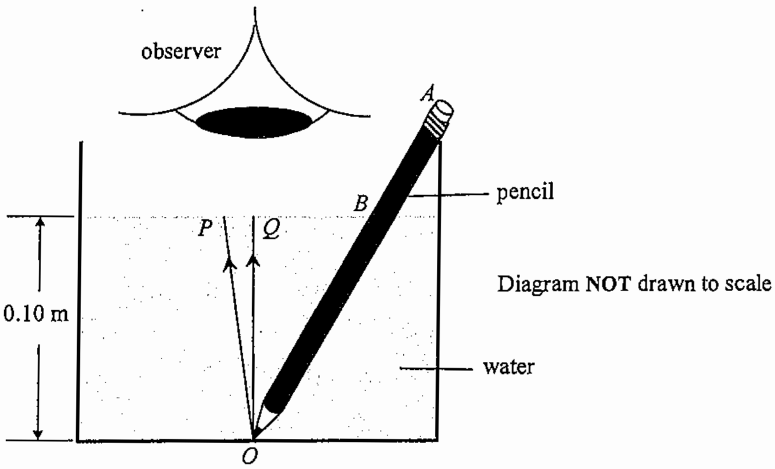
8. Figure 8.1 shows a pencil in a glass of water with an observer viewing vertically downwards. Given that the refractive index of water is 1.33.

Figure 8.1



Light rays P and Q come from the tip O of the pencil, with Q along the vertical.

(a) On Figure 8.1, complete the paths of these light rays and indicate the image of O seen by the observer (label this as I). (2 marks)

(b) The angle between ray P and ray Q is 1.5°. Find the angle of refraction of ray P. (2 marks)

(c) Hence, or otherwise, find the distance between image I and the water surface if the depth of water is 0.10 m. (2 marks)

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As shown in Figure 10.2, the rain sensor is composed of a light emitting diode (LED) and an infra-red sensor. The LED emits an infra-red signal and the infra-red sensor measures the intensity of the signal. When the intensity changes, the wipers are activated. An infra-red ray from the LED incident at surface A at 45°, and the infra-red ray finally received by the infra-red sensor are drawn in Figure 10.2. Assume surface A is parallel to surface B. It is given that the critical angle of the glass for infra-red is 44.0°.

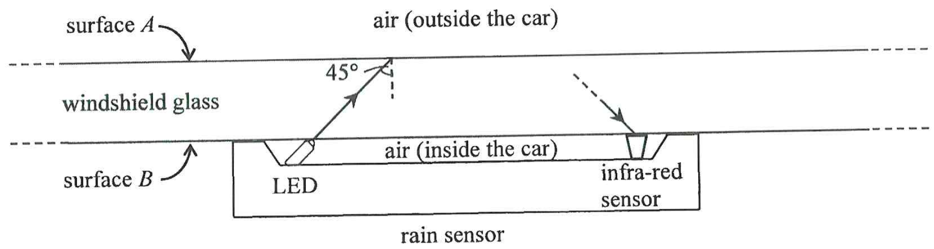

Figure 10.2

(a) Explain how the infra-red ray in Figure 10.2 travels from the LED to the infra-red sensor. (4 marks)

Answers written in the margins will not be marked.

Figure 10.3(a)

Figure 10.3(b)

(i) Given that the refractive index of water for the infra-red is 1.26, find the critical angle of the glass-water interface for the infra-red ray. (3 marks)

(ii) Sketch in Figure 10.3(b) how the infra-red ray propagates after being incident on the glass-water interface. (2 marks)

(iii) The wipers are activated when the intensity of the signal received by the infra-red sensor changes. Explain briefly how the intensity changes. (2 marks)

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